NTNU – Trondheim
Norwegian University of Science and Technology

TET4185: Power Markets, Resources and Environment
Hydro Generation

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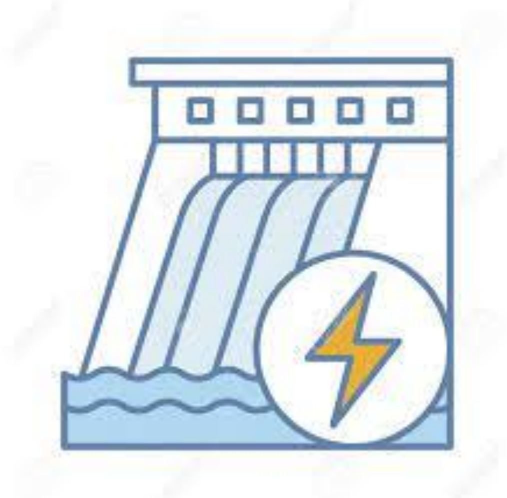


Overview

- Hydro: bathtub model without and with constraints

- Dealing with Uncertainty

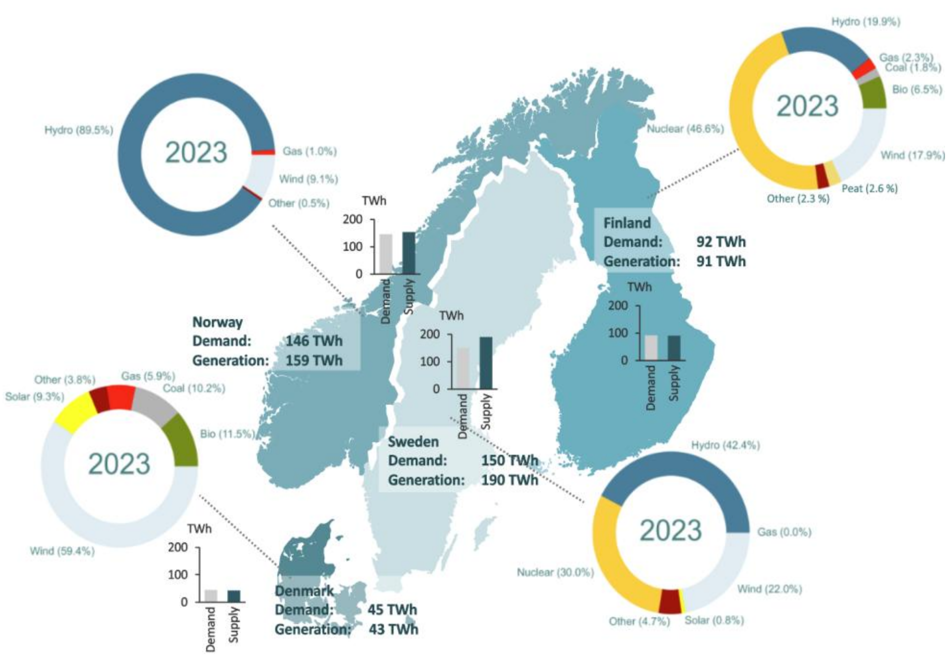
Learning Outcomes

- Analyze and apply hydropower dispatch principles

- Incorporate uncertainty in hydro generation planning

The Nordic power system is dominated by hydro, wind and nuclear power

Supply and demand of electricity in the Nordic power market



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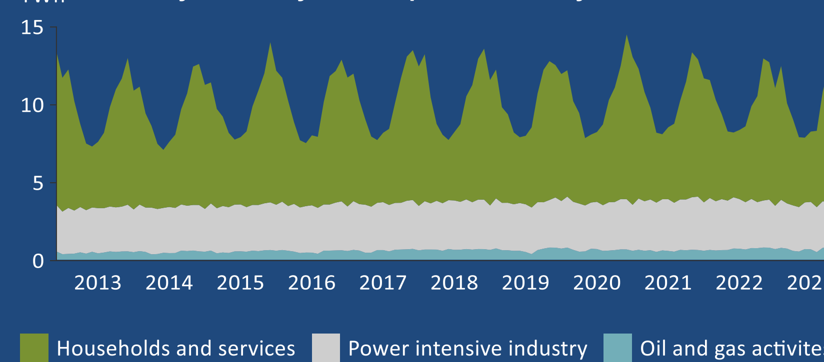
This makes the power system dependent on weather conditions

Weather dependency in both supply and demand makes the system vulnerable on its own

- Both wind power and hydropower is weather dependent – the production capacity depends on wind conditions and water inflow
- Water reservoirs, especially prevalent in Norway, reduce some unpredictability. Still, much hydropower relies on small or nonexistent reservoirs, and rainfall can vary significantly between years
- While hydro inflows are largest during the spring and summer, the demand is highest during winter. The electricity demand in Norway has strong seasonal variations, because of a large share of electrical heating.

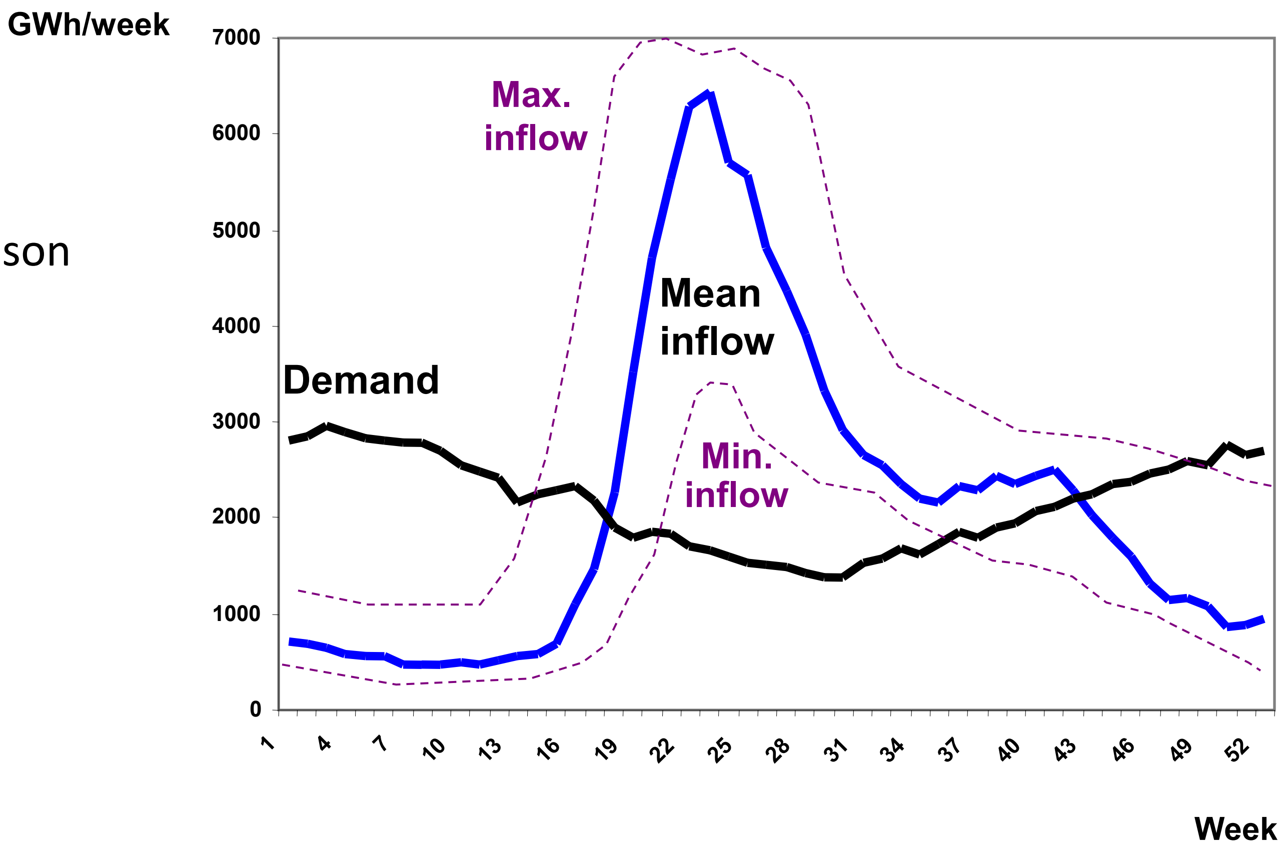
Monthly electricity consumption in Norway



THEMA Consulting Group

Weekly inflow and production of hydropower in Norway

- Filling Season
- Depletion Season



Hydro units

- The variable cost in hydro plant is almost zero
- The amount of water is limited
- There is limited storage capacity in the reservoir
- The only variable cost in using water can be represented in the form of an *opportunity* cost
- The cost today is the potential benefit obtained by using water tomorrow.

Market Clearing Without Demand Elasticity

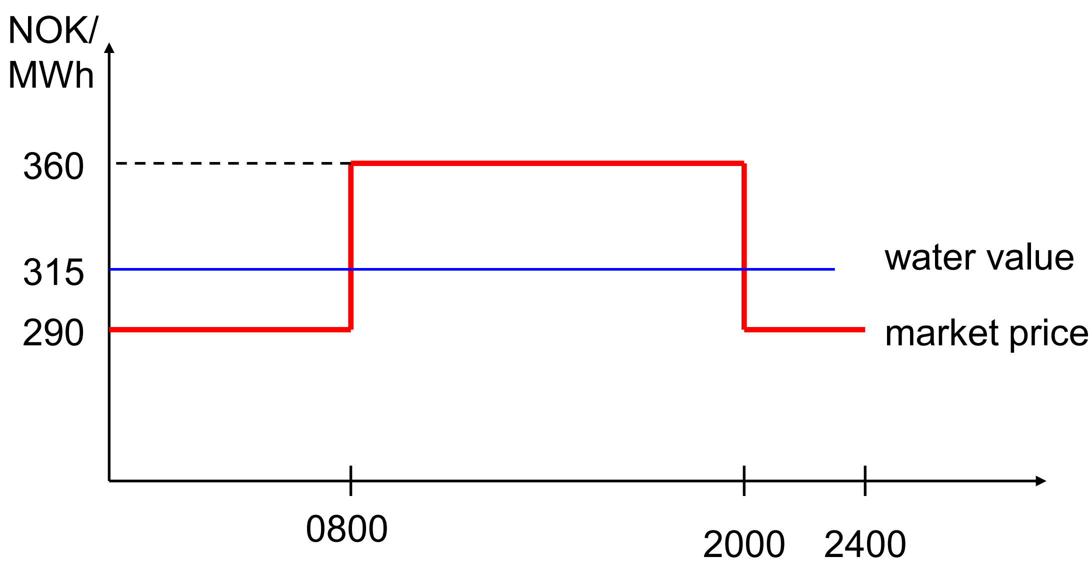
- Goal: Minimize operating costs
- Cost types
  - Fuel costs
  - Emission taxes
- Constraints
  - Generation = Total load + Losses
  - Max/min production per unit

Optimal Hydropower Dispatch

- For thermal plants we have assumed known fuel costs
- What about hydropower?
  - Water is free, but as a limited resource it has an alternative value
  - Reservoirs enable producers to save water at a time when prices are higher than today
- To determine when to use the water, it is therefore necessary to calculate its <u>expected future value</u>
  - The ***water value*** is the ***expected marginal value*** of having an additional unit of water in the reservoir
- If we know this value, the problem to solve is in principle the same as the problem for thermal units

Optimal Hydropower Dispatch– example

- Water value: 315 NOK/MWh
- Plant capacity 100 MW
- Market price:



Optimal Hydropower Dispatch– example

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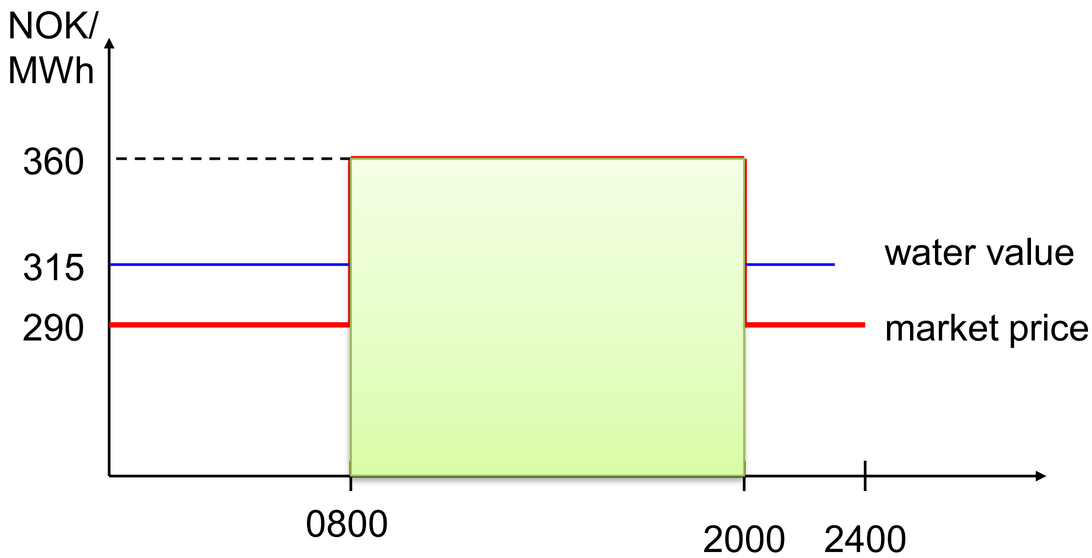
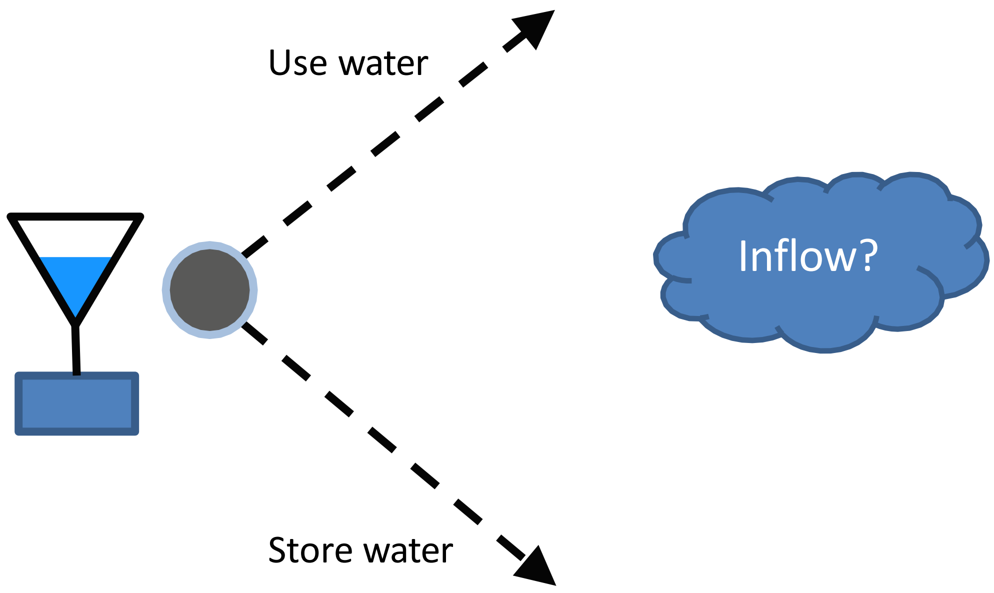
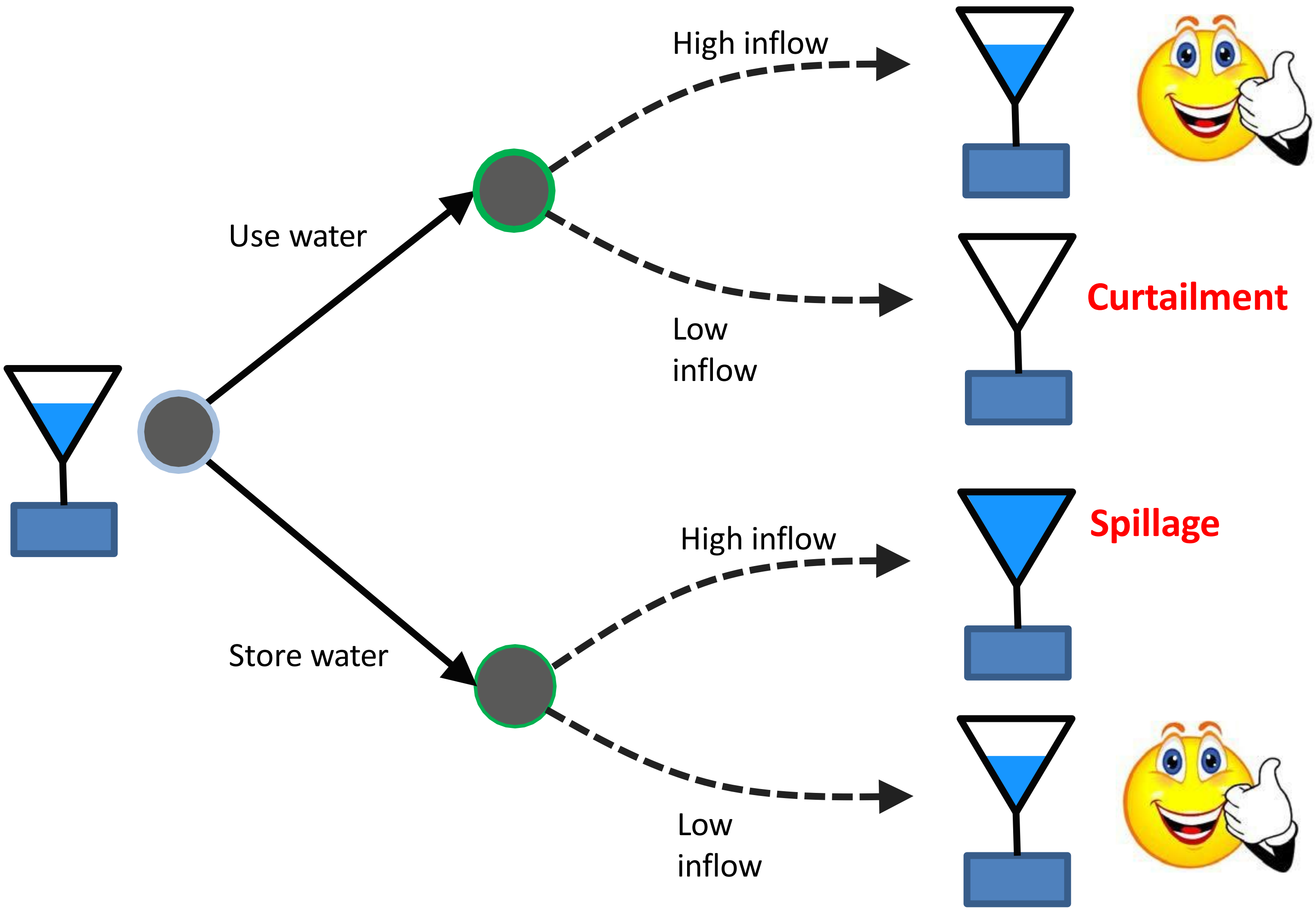


Hydropower planning consequences
(Basis for stochastic optimisation)



Hydropower planning consequences

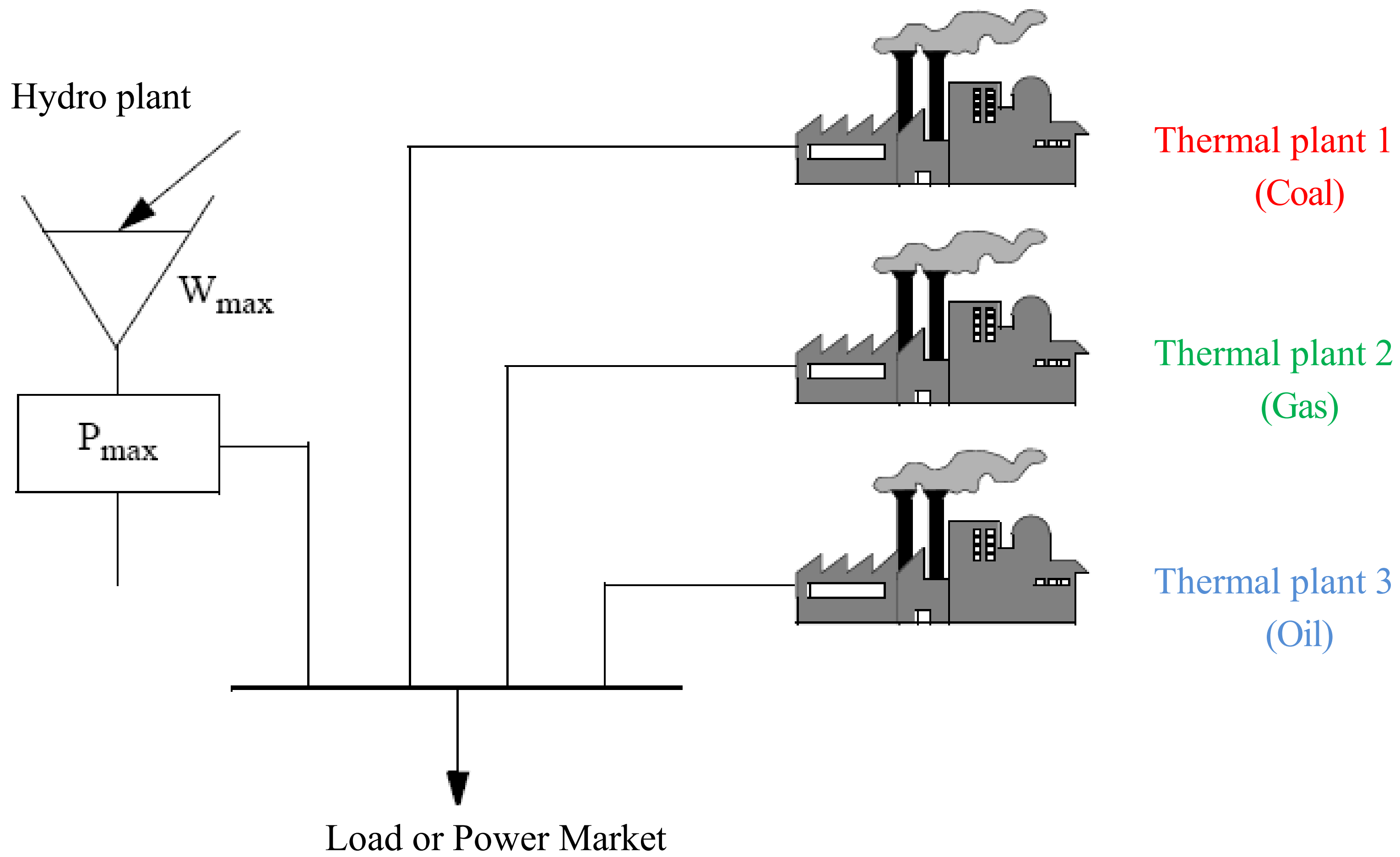


Water value – Definition

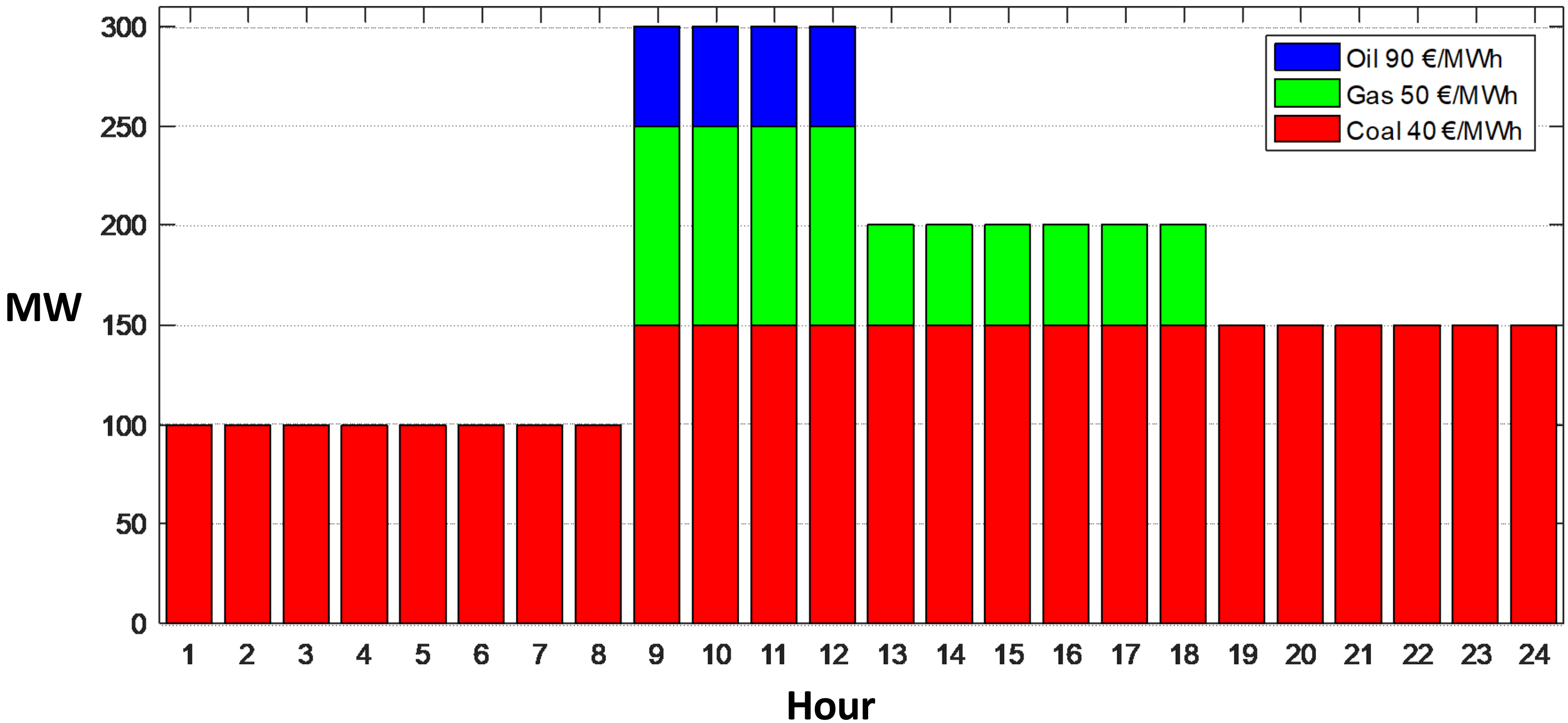
The ***water value*** is the ***expected marginal value*** of having an additional unit of water in the reservoir

We will use Stochastic Dynamic Programming to find the water value

Deterministic Example hydro-thermal system

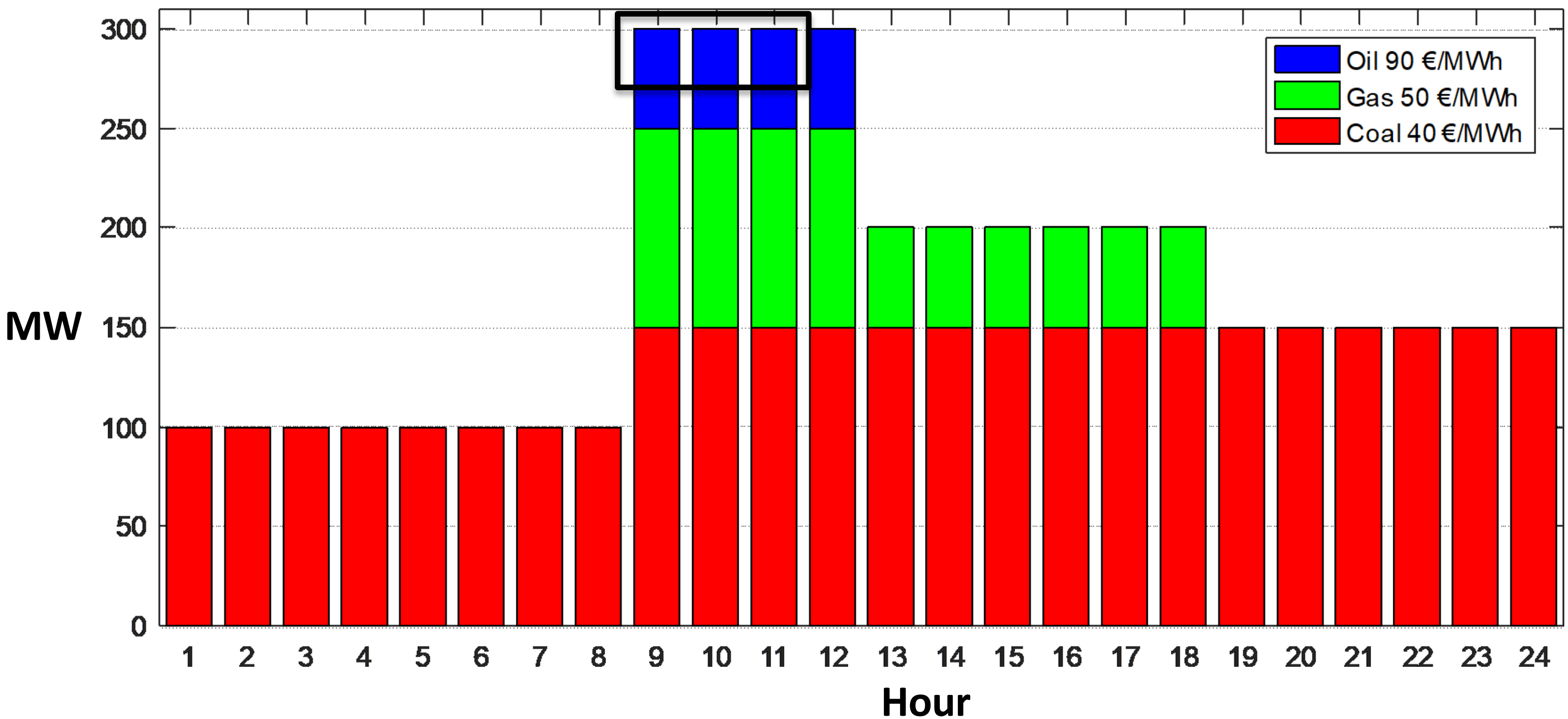


Water Value Calculation-Example



- Hydro: 25 MW, 75 MWh
  - How to use it optimally?
  - Marginal value of having 1 more MWh?

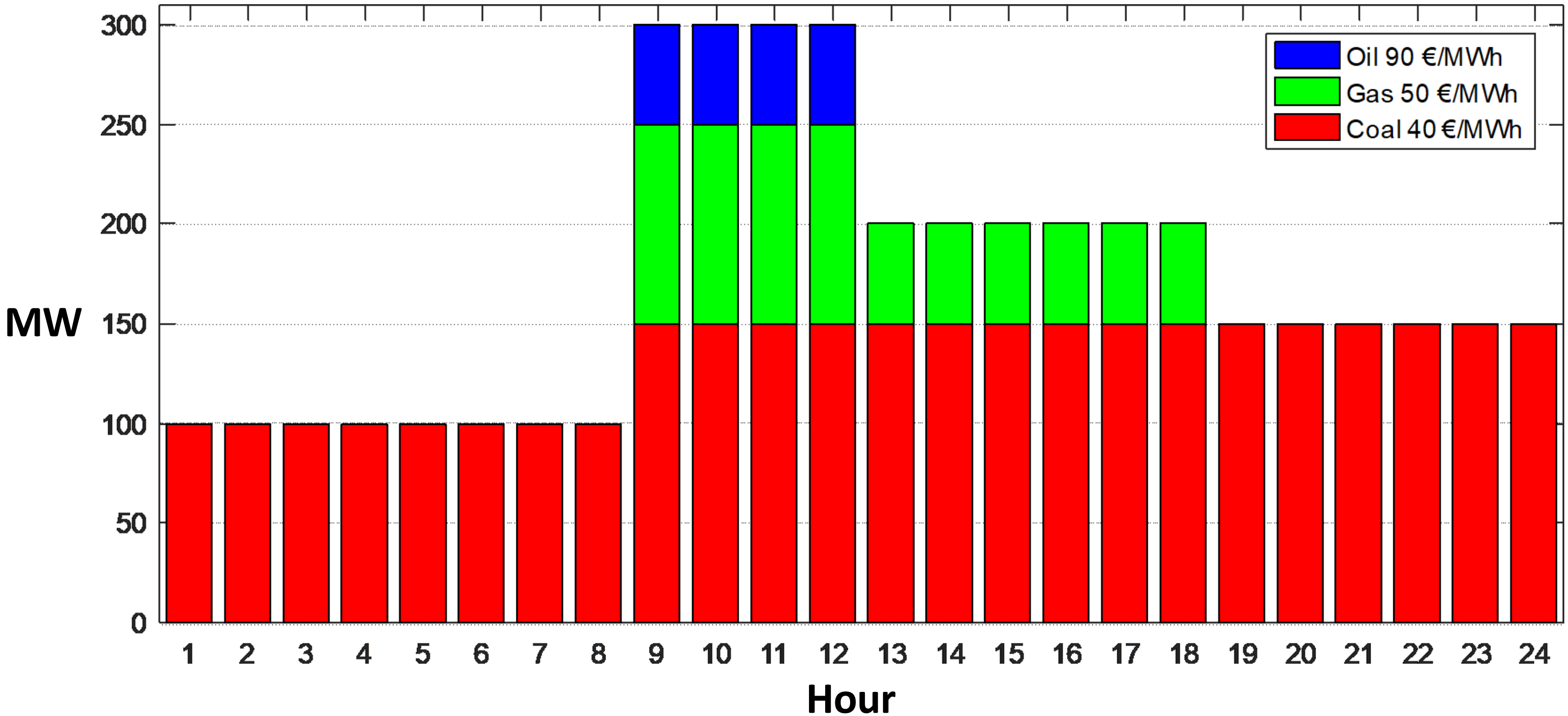
Water Value Calculation-Example (1)



- Hydro: 25 MW, 75 MWh
  - How to use it optimally?
  - Marginal value of having 1 more MWh?

Hydro replaces Oil -> Water value is 90 €/MWh

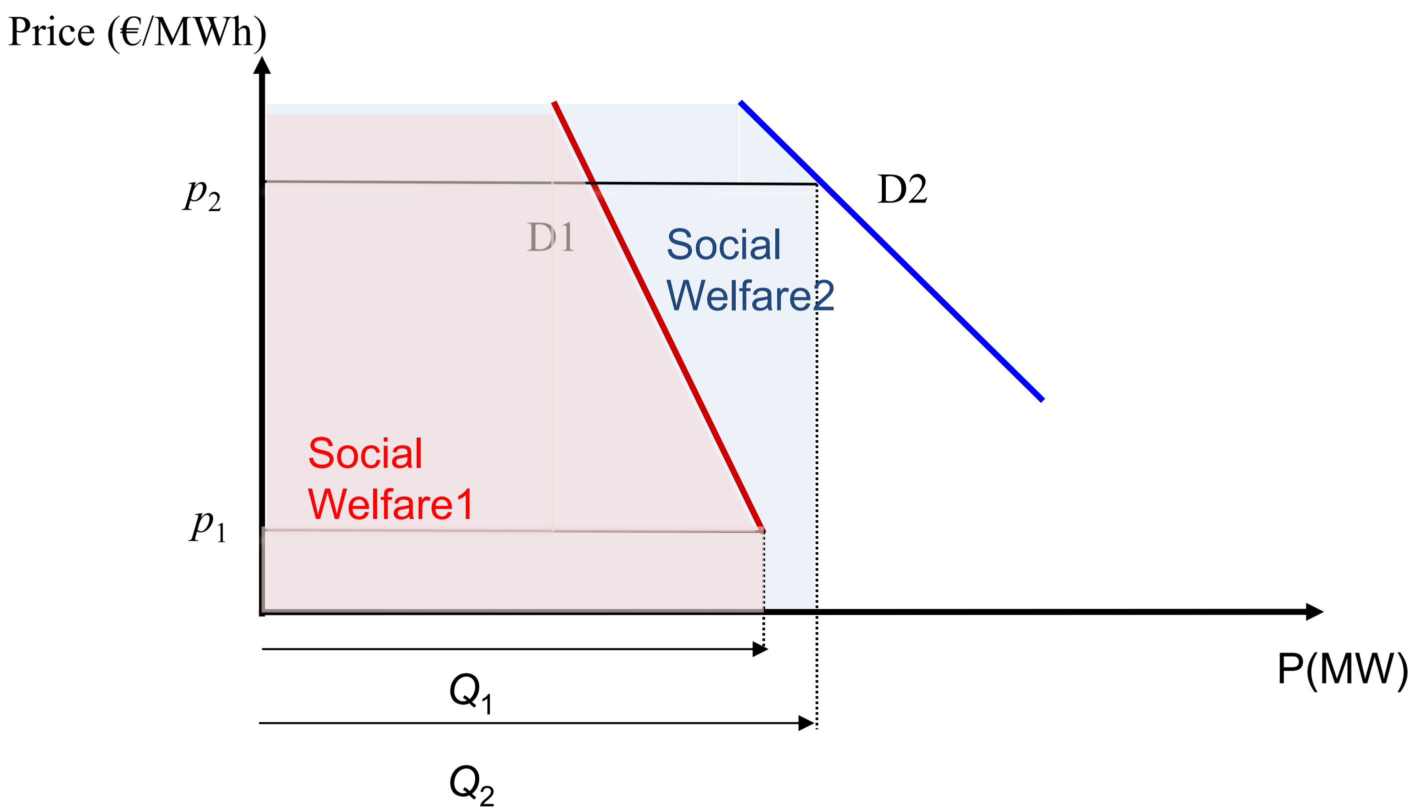
Water Value Calculation-Example (2)



- Find the water value of the following hydro power specifications:
  - A) 25 MW power capacity and 100 MWh stored energy 50
  - B) 25 MW power capacity and 300 MWh stored energy 40
  - C) 100 MW power capacity and 150 MWh stored energy 90

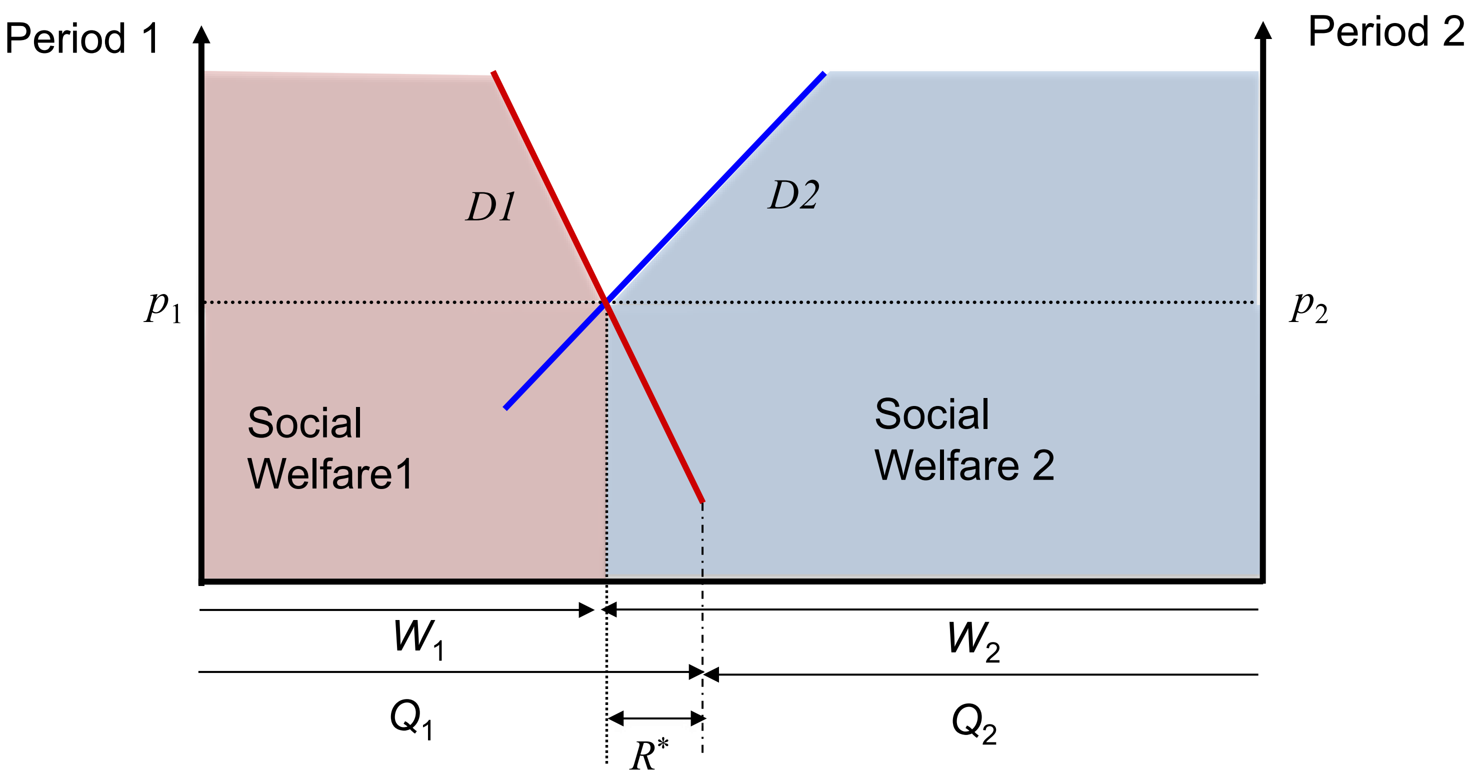
BATHTUB MODEL

Bathtub model-**Deterministic** situation



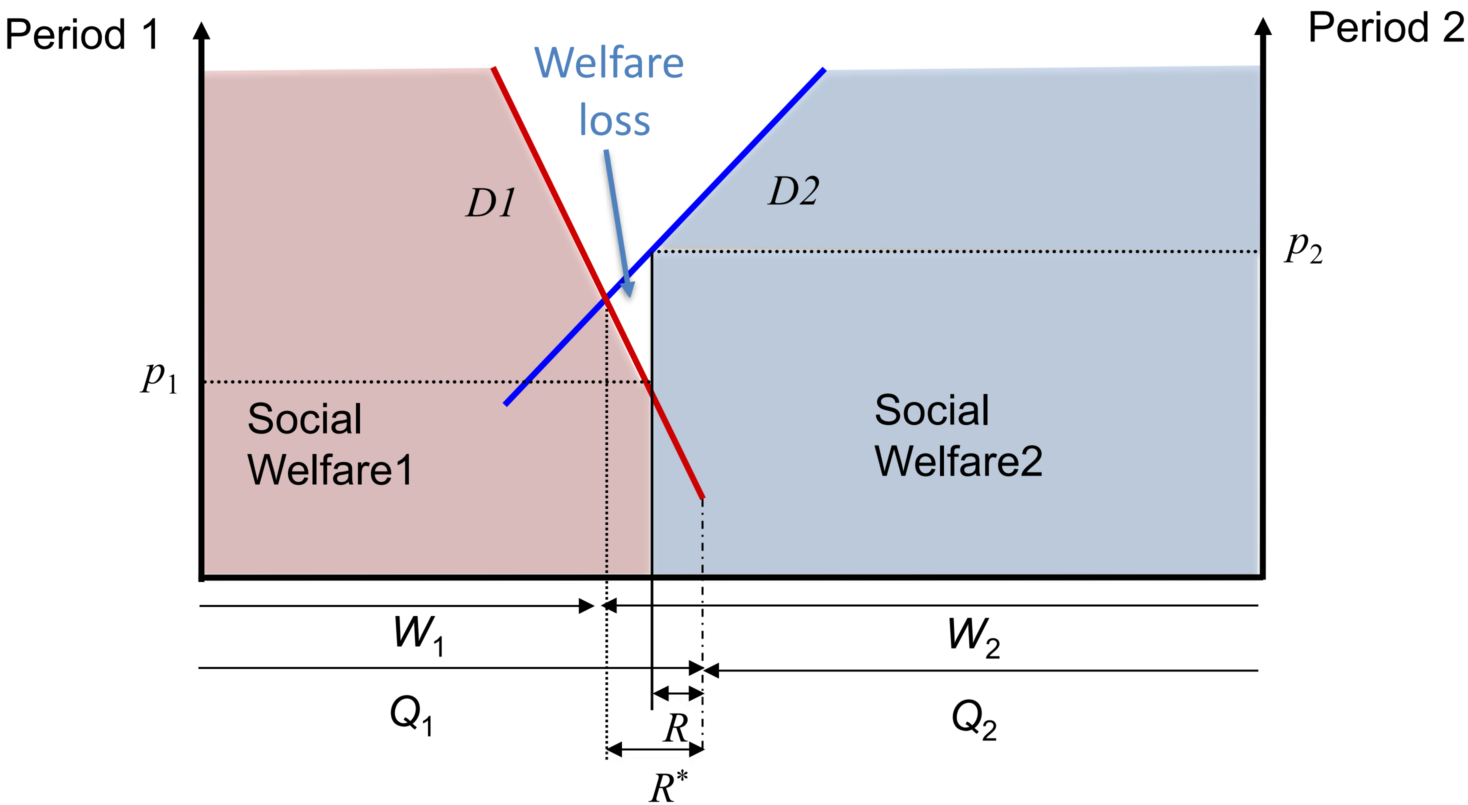
Bathtub model-Deterministic situation

Equal prices in each period



Bathtub model with reservoir constraint

Different prices in each period caused by reservoir constraint

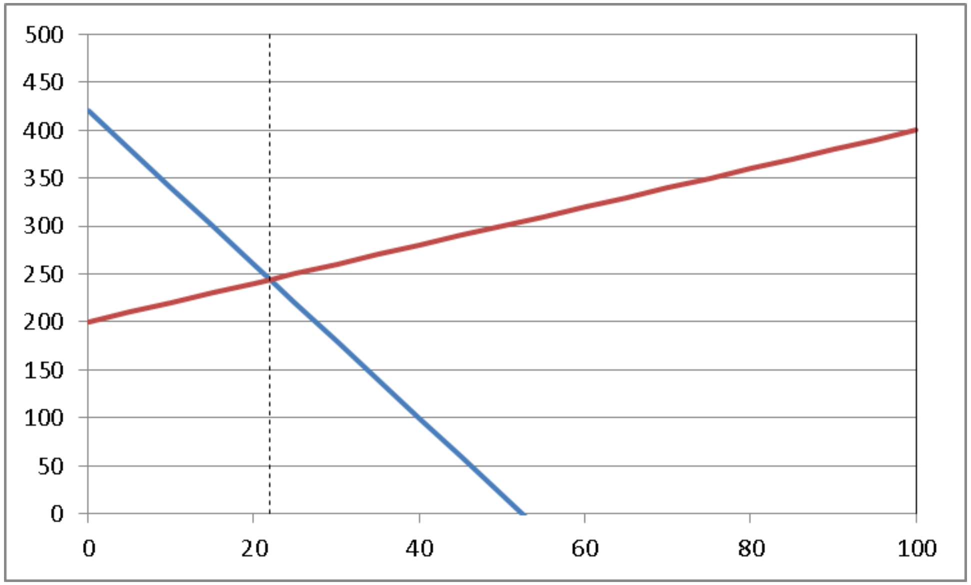


Bathtub model example!

Bathtub model

$p_1 = 420 - 8W_1$
$p_2 = 400 - 2W_2$
(p_2 from right side)

$W_1 + W_2 = 100$
$Q_1 = 60$, $Q_2 = 40$, $R = \infty$



Q1: find optimal price and quantities

Q2: $R = 30$, find optimal price and quantities

Q3: $R = \infty$, $W_{2,max} = 60$, find optimal price and quantities

Solutions

Q1: No constraint

$420 - 8W_1 = 400 - 2W_2 = 400 - 2(100\text{-}W_1)$

$\rightarrow W_1 = 22,\ W_2 = 78,\ p_1 = p_2 = 244$

required R: 60 – 22 = 38

Q2: Constrained Reservoir capacity $\rightarrow$ R = 30

$W_1 = 60 - 30 = 30,\ W_2 = 100 - 30 = 70$

$p_1 = 420 - 8 \times 30 = 180,\ p_2 = 400 - 2 \times 70 = 260$

Q3: Constrained generation capacity $\rightarrow W_{2,max} = 60$

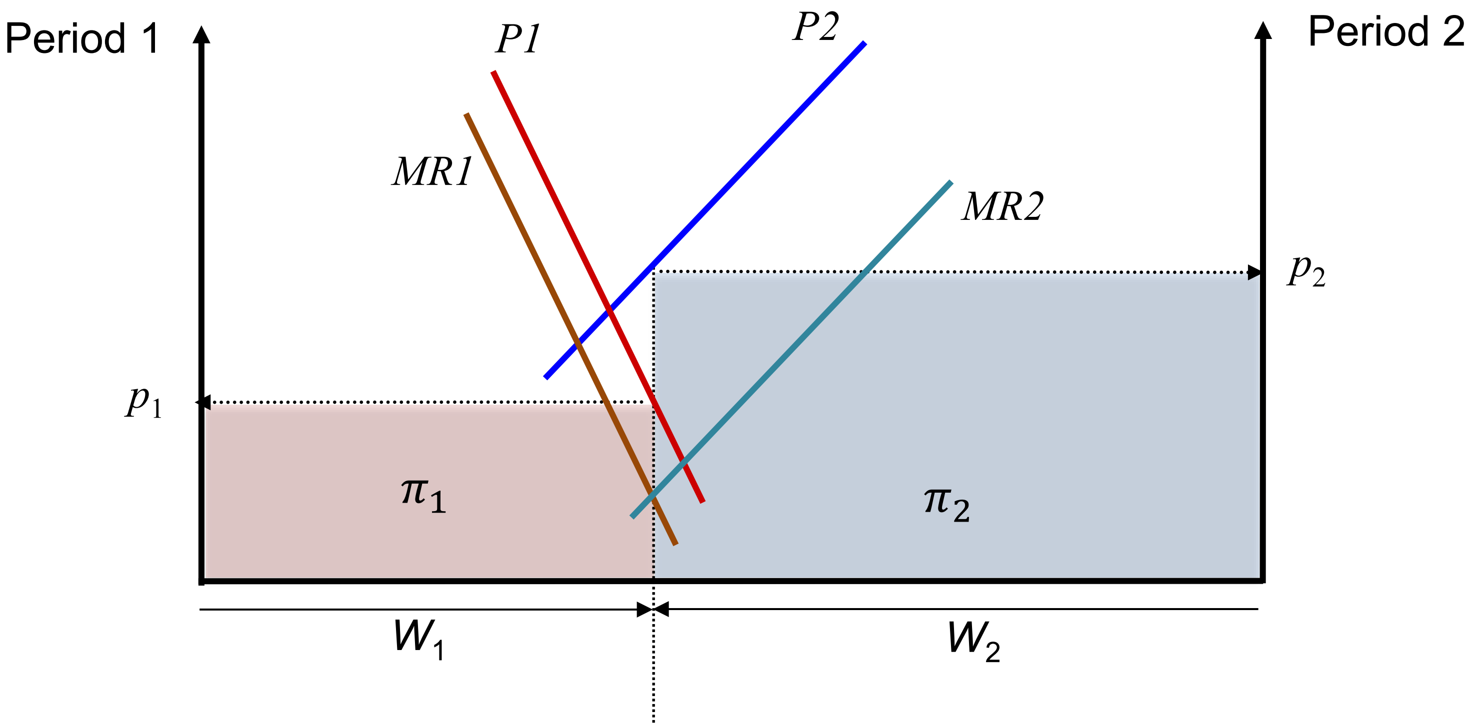
$W_2 = W_{2,max} = 60,\ W_1 = 40 \quad p_1 = 100,\ p_2 = 160$

Lessons from bathtub model

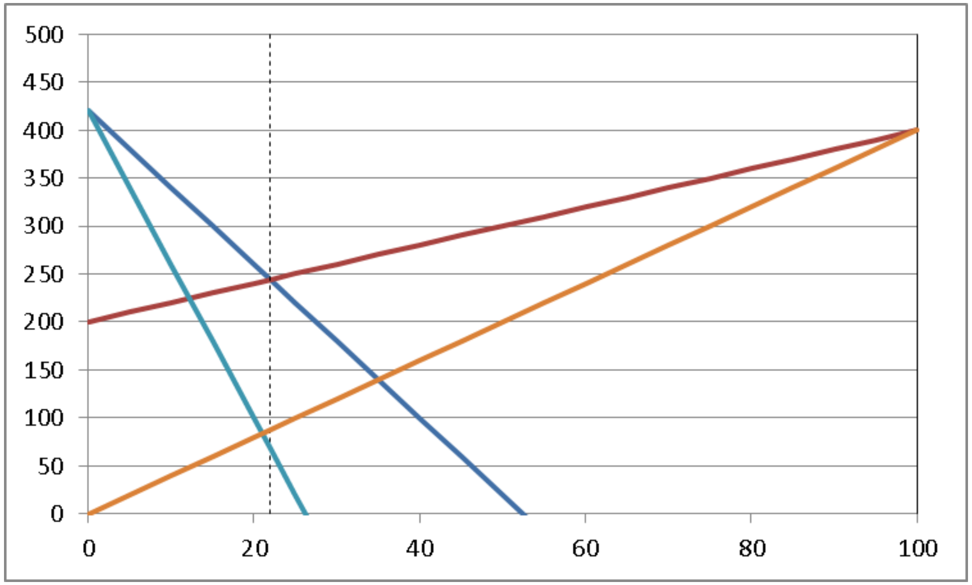
- With infinite reservoir and production capacities, prices will always be equal
  - Norwegian system: relatively small price differences

- Constraints lead to price differences
  - reservoir capacity: not possible to transfer enough water between periods
  - generation capacity: not possible to produce enough according to theoretical optimum
  - additional constraint in real systems?
    - transmission capacity!

Monopoly



Monopoly example!



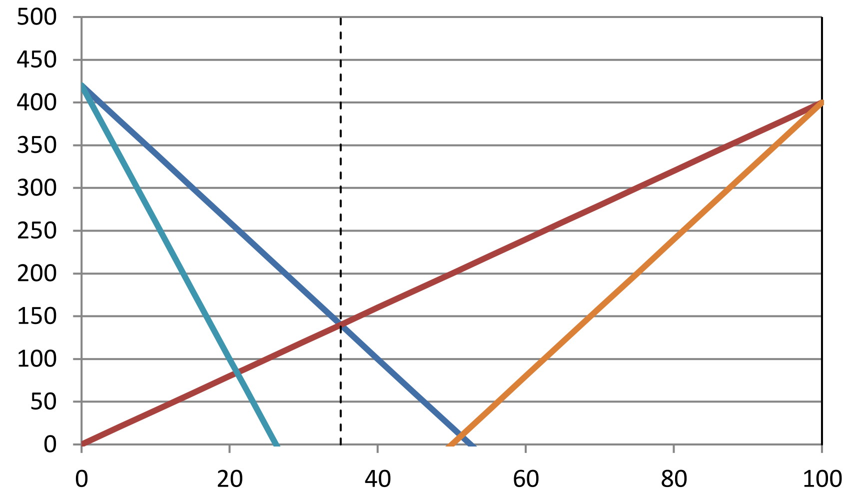
- Linear curves, marginal revenue: $MR = a - 2bw \rightarrow \begin{cases} MR_1 = 420 - 16W_1 \\ MR_2 = 400 - 4W_2 \end{cases}$
- Equal MR yields:
  $W_1 + W_2 = 100 \rightarrow W_1 = 21,\ W_2 = 79$
  $p_1 = 420 - 8W_1 \rightarrow p_1 = 420 - 8\times21 = 252$
  $p_2 = 400 - 2W_2 \rightarrow p_2 = 400 - 2\times79 = 242$

Monopoly: slightly different model

Monopoly

p_1 = 420 – 8W_1
p_2 = 400 – 4W_2
W_1 + W_2 = 100

Perfect competition:
W_1 = 35, W_2 = 65, p=140
Revenue: 14000



- Equal *MR* yields:
  W_1 = 34.2, W_2 = 65.8, MR_1 and MR_2 <0!
  p_1 = 146.4, p_2 = 136.8, Revenue: 14008.3
  this assumes the monopolist must utilize all water!
- If possible to spill water, we utilize like earlier *MR*=*MC*=0:
  W_1 = 26.25, W_2 = 50, *spill* = 100 – 26.3 – 50 = 23.7
  p_1 = 210, p_2 = 200, *Revenue* = 15512.5

Monopoly and hydro power

- If the monopoly producer can spill water, he/she can significantly increase prices and his profits

- If this is forbidden and there is sufficient control, the effect of abusing monopoly power is only marginal
  - because withholding in one period increases production in other periods

Scheduling Toolchain

Long-term
EMPS
(SDP,aggregation/disaggregation)

↓

Price forecast

↓

Medium-term
ProdRisk
(SDP/SDDP)

↓

Water values

↓

Short-term
SHOP
(MIP/SLP)

Horizon

- ✓ Multi-area model
- ✓ Internalized market
- ✓ Stochastic inflow and demand
- ✓ Minimize cost (maximize socio-economic surplus)

Several years, weekly decision stages

- ✓ Single producer, single bus bar
- ✓ External market
- ✓ Stochastic inflow and price
- ✓ Maximize profit

Several years, weekly decision stages

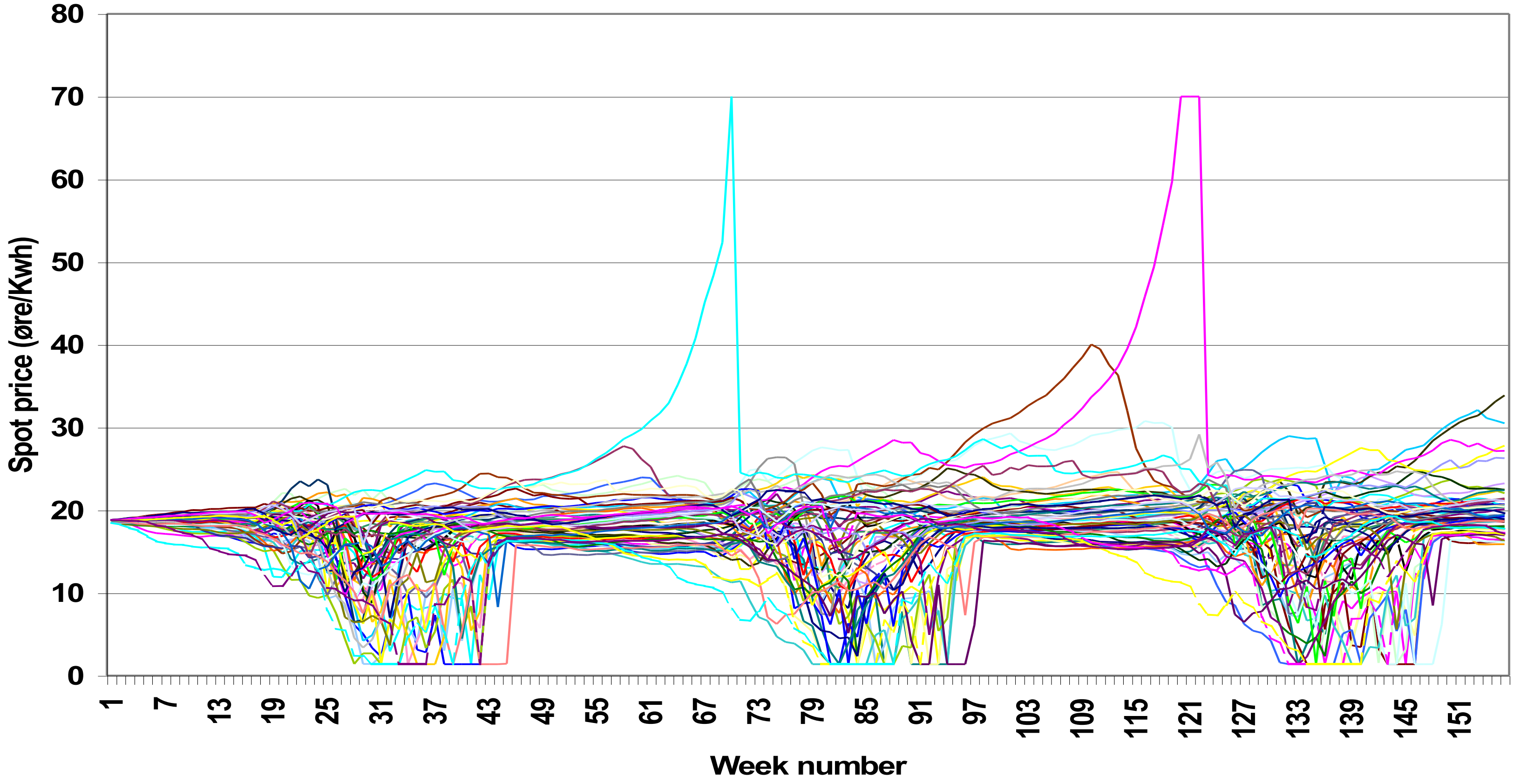
- ✓ Single producer, single bus bar
- ✓ External market
- ✓ Deterministic
- ✓ Maximize profit

One-Two weeks

A. Helseth and A. C. Geber de Melo, “Scheduling Toolchains in HydroDominated Systems: Evolution, Current Status and Future Challenges for Norway and Brazil,” SINTEF Energy Research, Tech. Rep., 2020, https://sintef.brage.unit.no/sintef-xmlui/handle/11250/2672581?locale-attribute=en .

Dealing with **Uncertainty** in input data
Price scenarios

- A sample 3-year (156 week) price forecast for approximately 70 scenarios



Stochastic market prices

- **We have hydro plant with reservoir with capacity of 100 GWh, the current price in the market is 100 €/MWh (first-stage, S1)**
- Shall I produce power or wait until the next time step (Second-stage,S2)?
- Recall that we don't know the actual market price realization
- We only have some price forecasts!
  150 €/MW?
  100 €/ MW?
  50 €/ MW?
  0 €/ MW?
- Which forecast should be considered as the future price in the market?
  - Average forecast?
  - Any other value?

Stochastic market prices

- **<u>In first stage*:</u>**

Isn’t it smarter to consider each price forecast as a potential “scenario” in the second stage, evaluate dispatch consequences under every scenario, and then determine first stage dispatch according to all scenarios?

first stage (S1)

second stage (S2)

S1 dispatch should be determined based on all S2 scenarios!

A weight for each scenario!

$\xrightarrow{0.25}$ S2 scenario 1: Market price is 150 €/MW

$\xrightarrow{0.25}$ S2 scenario 2: Market price is 100 €/MW

$\xrightarrow{0.25}$ S2 scenario 3: Market price is 50 €/MW

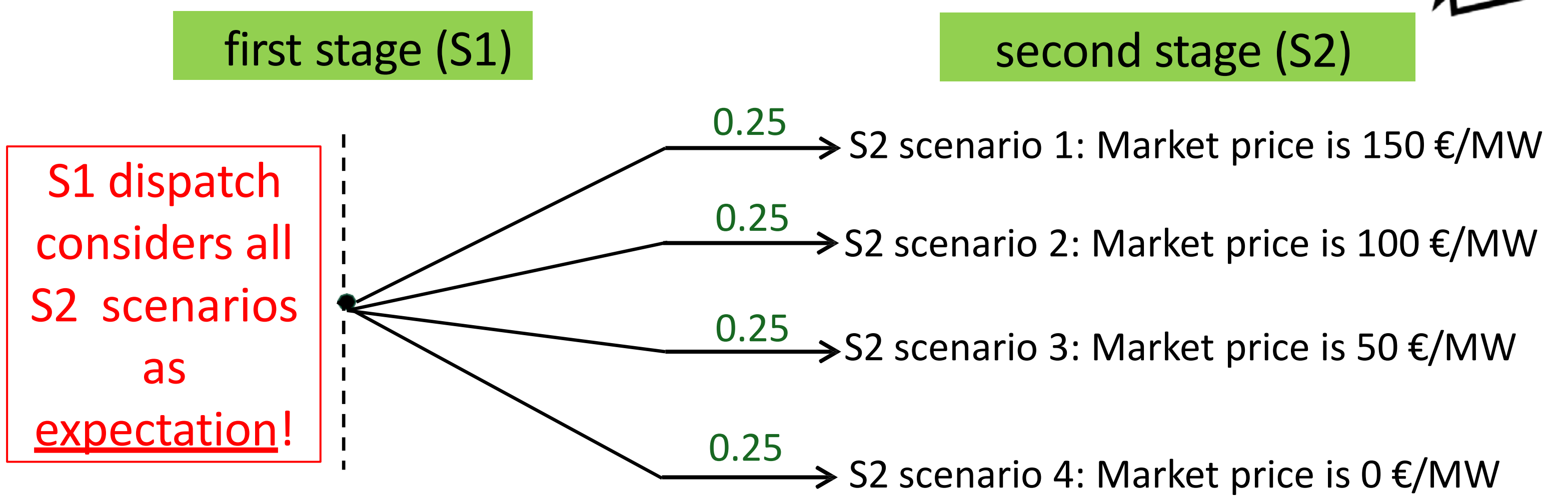
$\xrightarrow{0.25}$ S2 scenario 4: Market price is 0 €/MW

*Each stage represents a point in time where decisions are made, i.e., when new information is received.

Stochastic market prices

- **In first stage:**

Isn’t it smarter to consider each price forecast as a potential “scenario” in the second stage, evaluate dispatch consequences under every scenario, and then determine first stage dispatch according to all scenarios?



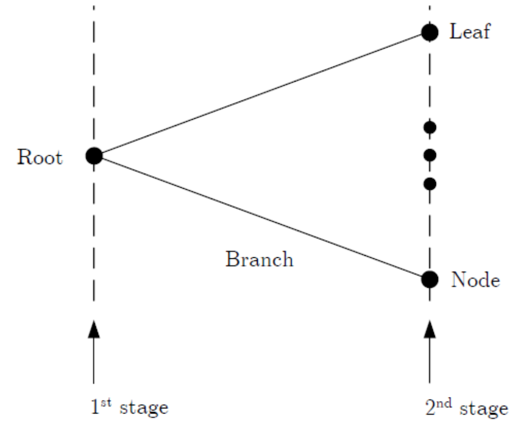
*recourse** decisions

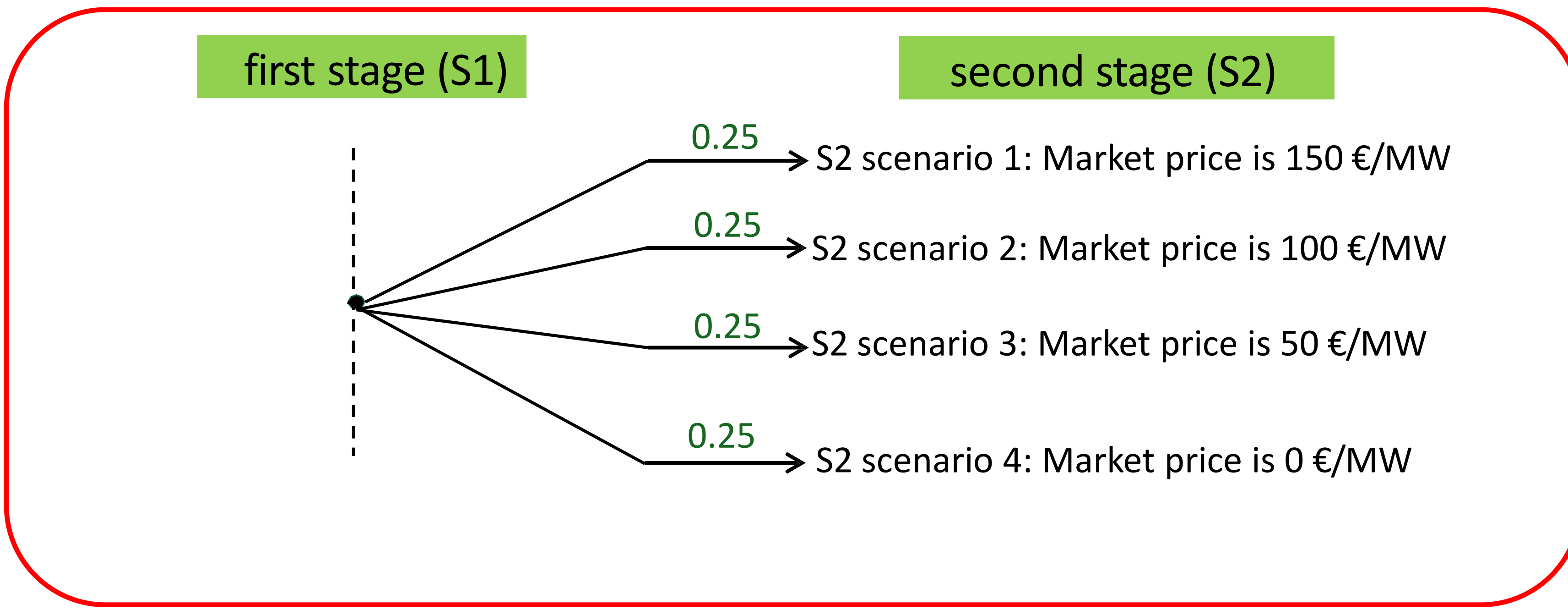
* The term *recourse* points to the fact that second-stage decisions enable the decision-maker to adapt to the actual outcomes of the random events.

Two-stage stochastic Problem

Scenario tree for two-stage problem

- First-stage or here-and-now decisions
- Second-stage or wait-and-see decisions.





Two-stage Programming

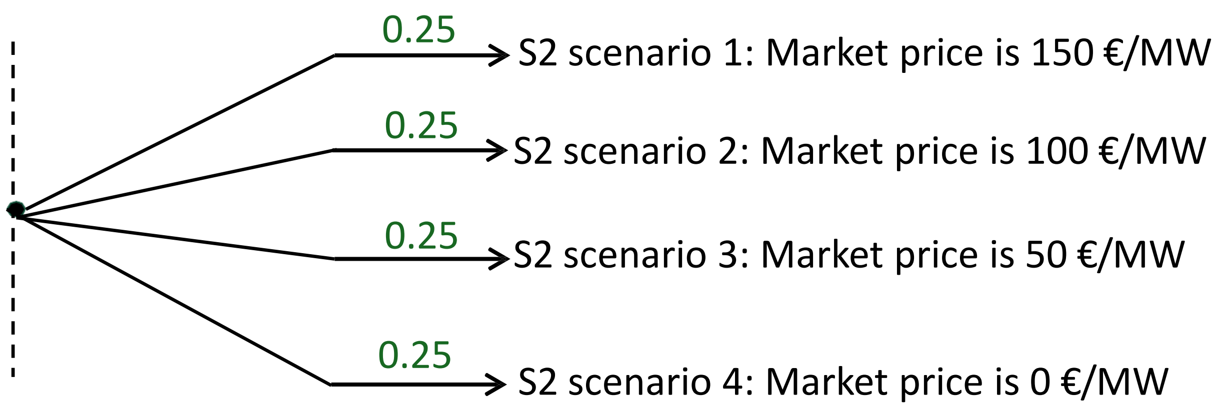
- **<u>In first stage:</u>**

Look-ahead strategy

Stochastic hydro dispatch

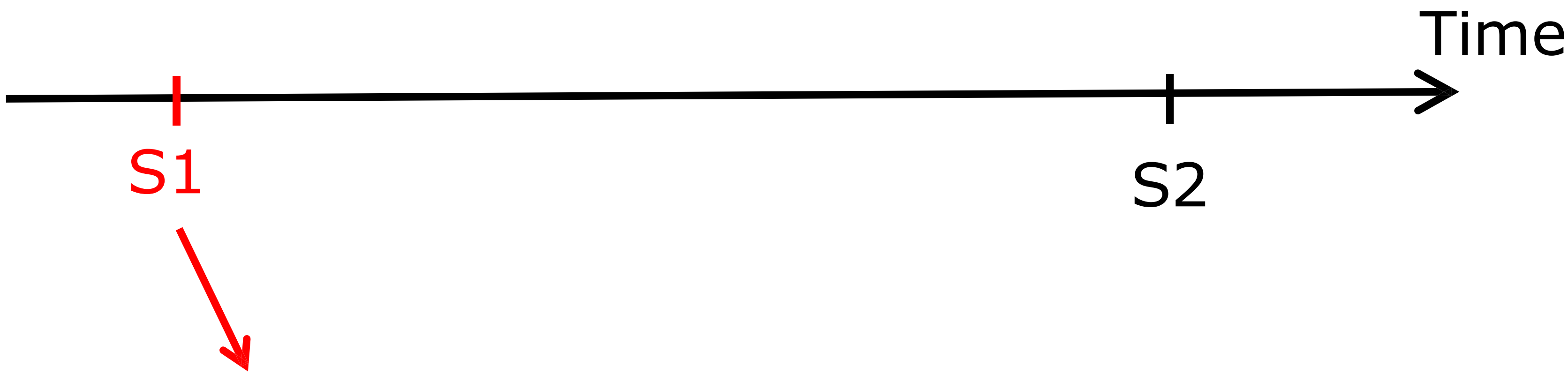
first stage (S1) second stage (S2)

S1 dispatch considers all S2 scenarios as <u>expectation</u>!

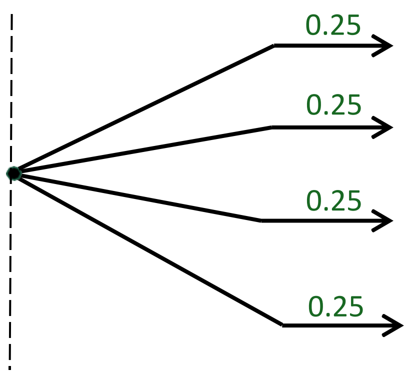


Hydro dispatch

Using **stochastic** market prices:



Dispatch based on a future valuation of stored hydro, e.g.,



S2 scenario 1: Market price is 150 €/MW

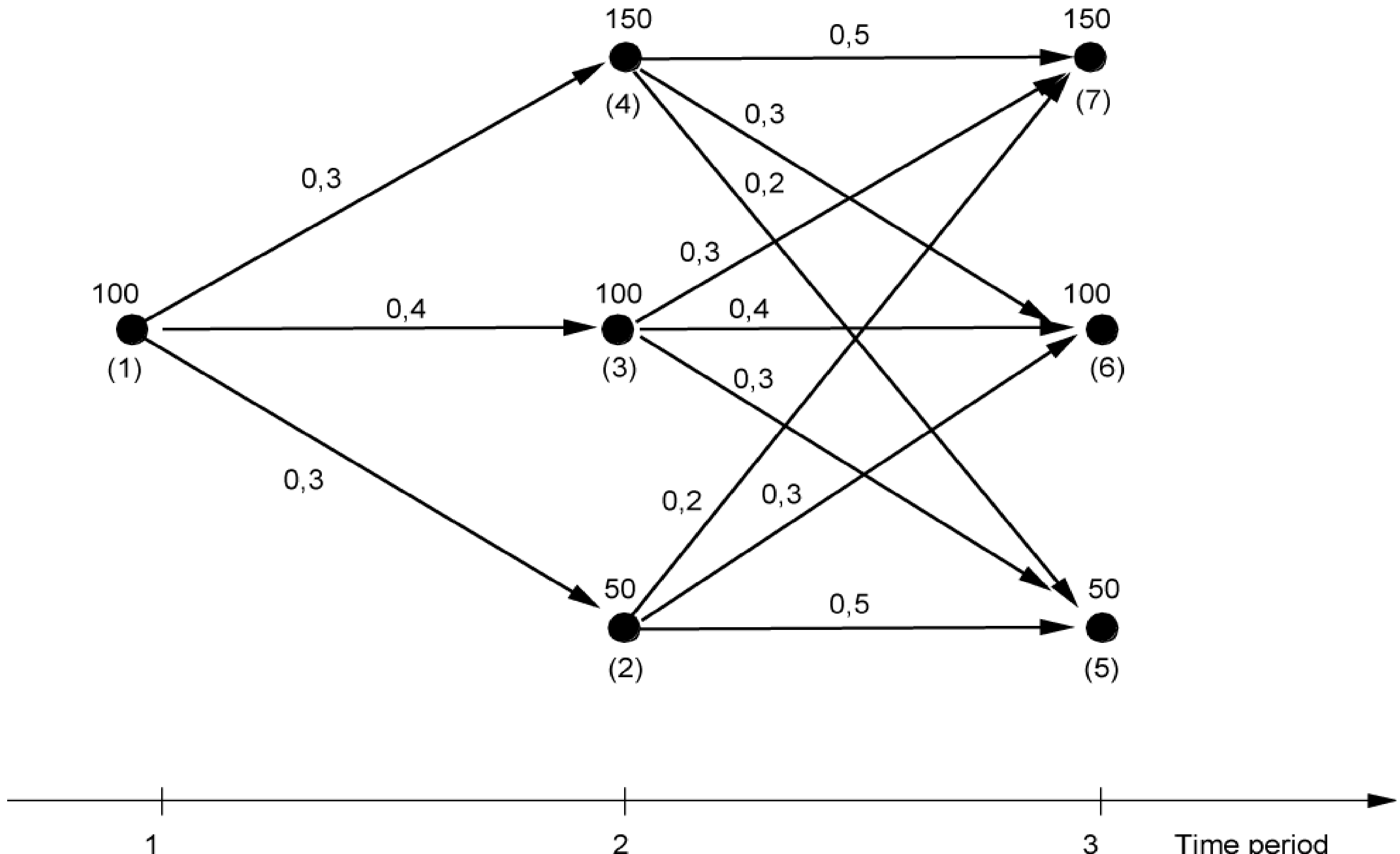
S2 scenario 2: Market price is 100 €/MW

S2 scenario 3: Market price is 50 €/MW

S2 scenario 4: Market price is 0 €/MW

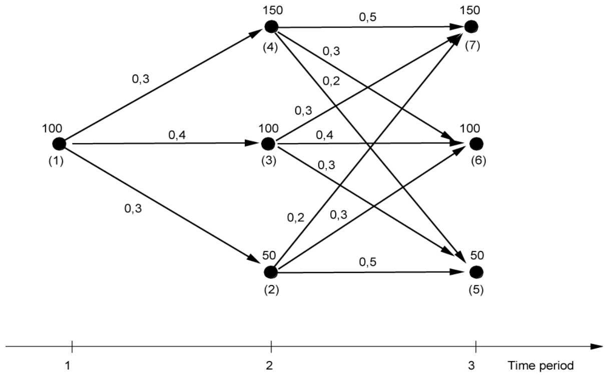
Example: “Recombining” Scenario tree for possible future market price in a multistage problem

We have hydro plant with reservoir capacity of 100 GWh

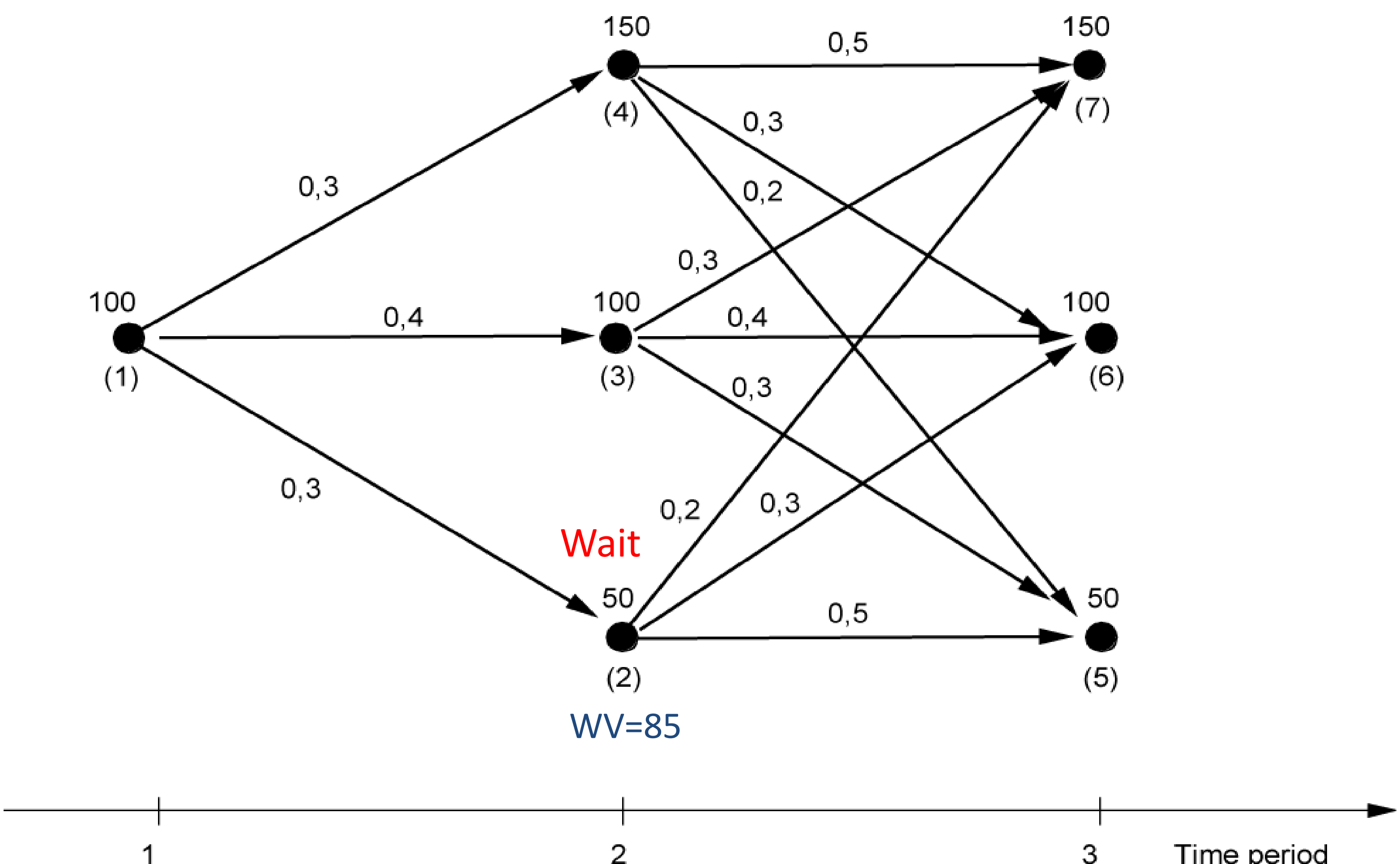


Stage 2-Node 2

Node	Water Value (expected future value of water)	Market price	Production/ sales
(2)	prob: 0.2 → 150 0.3 → 100 0.5 → 50 WV = 0.2 *50 + 0.3*100 + 0.5* 50 = 85	50	0

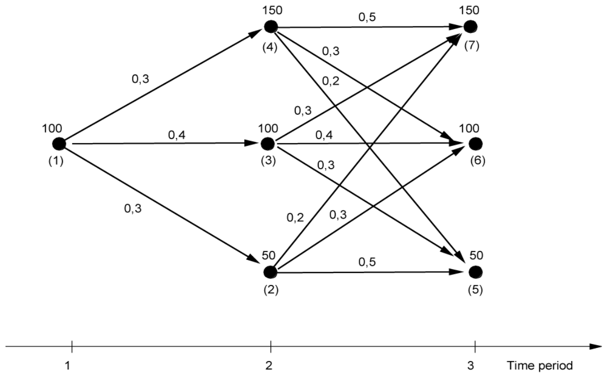


Results: Optimum strategy (Stage 2-Node 2)

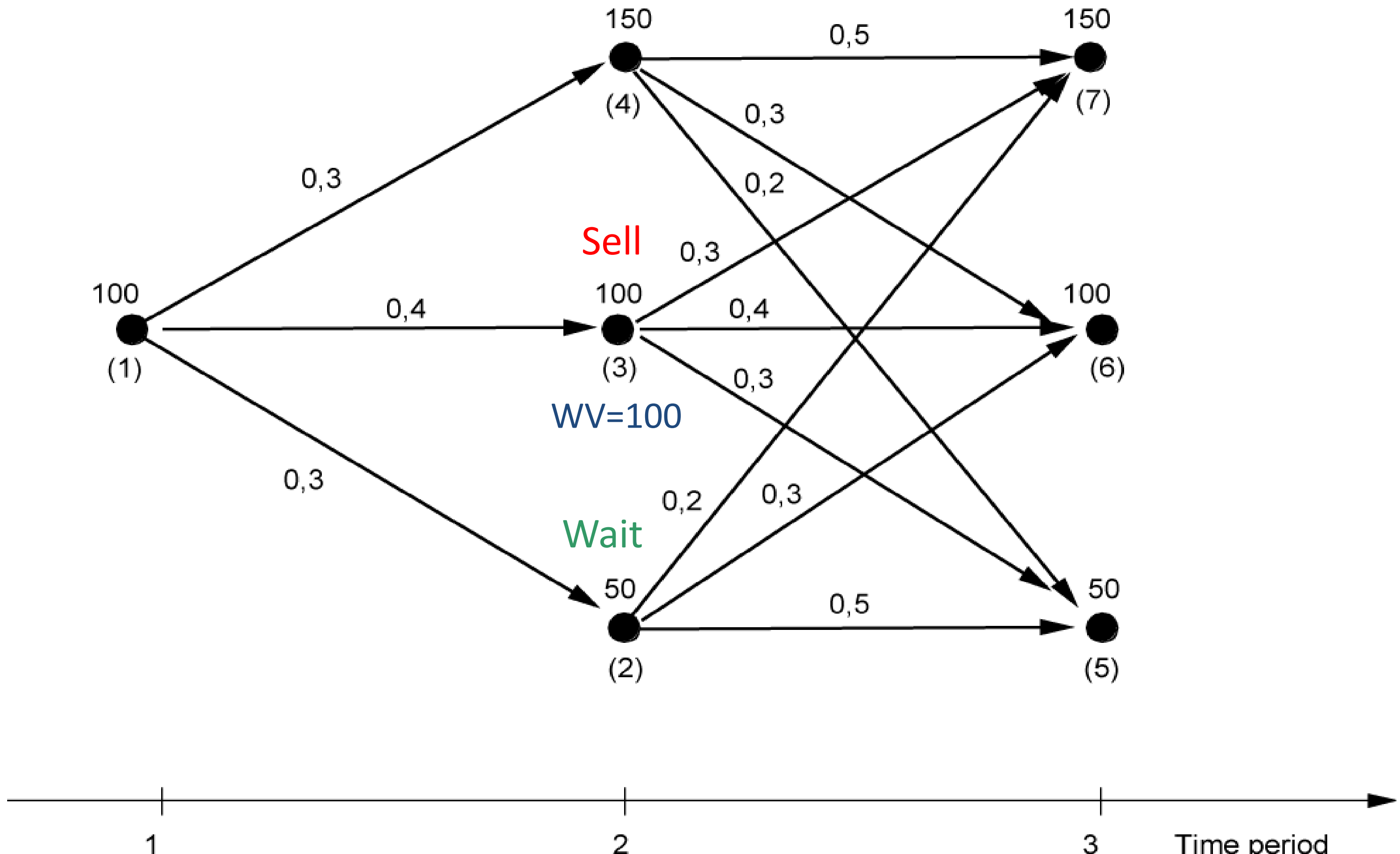


Stage 2-Node 3

Node	Water Value (expected future value of water)	Market price	Production/ sales
(3)	prob: 0.3 → 150 0.4 → 100 0.3 → 50 WV = 0.3 *150 + 0.4*100 + 0.3* 50 = 100	100	0 or 100

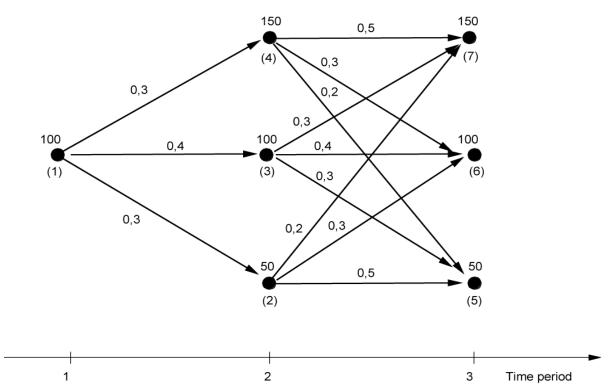


Results: Optimum strategy (Stage 2-Node 3)

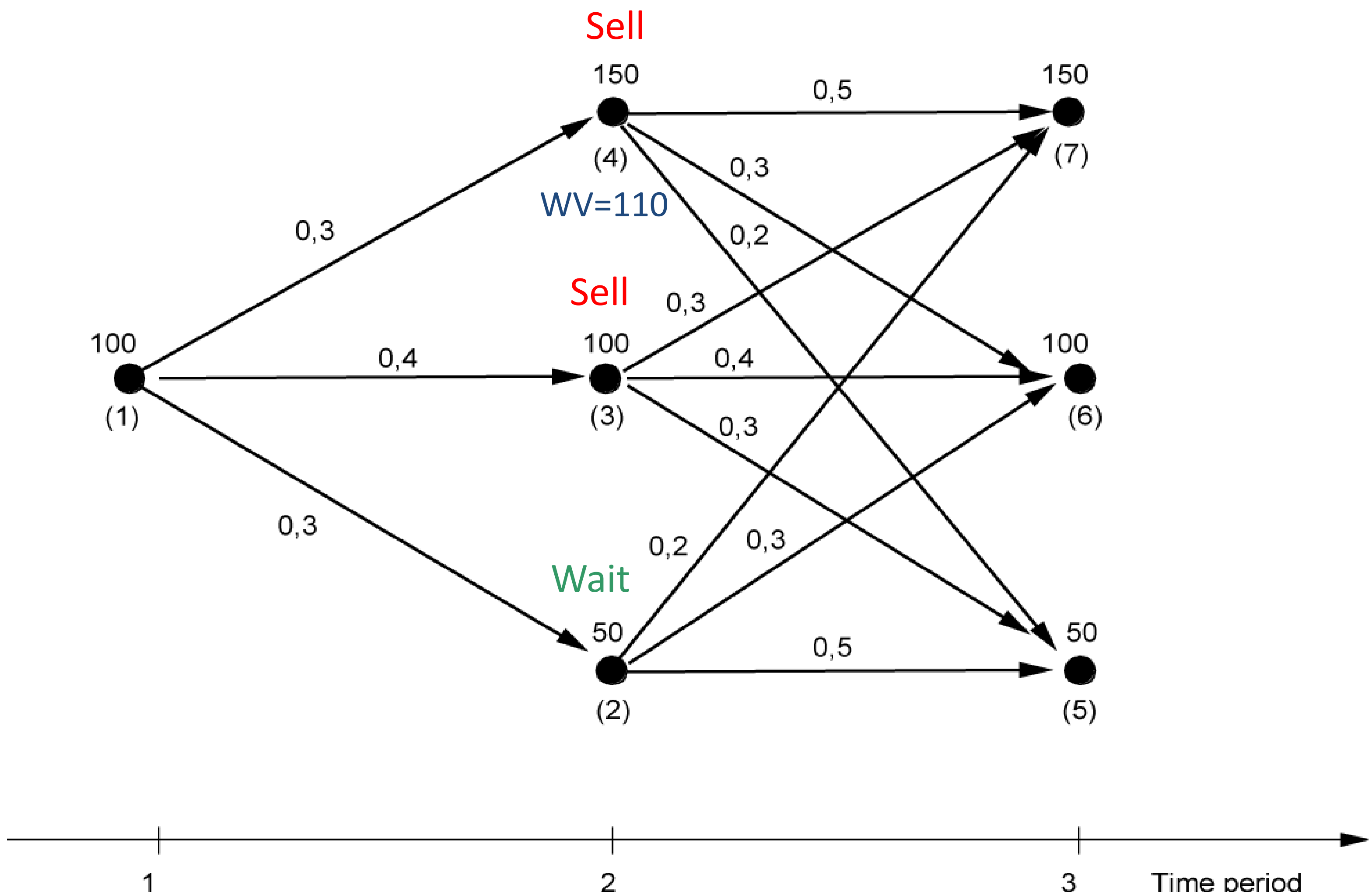


Stage 2-Node 4

Node	Water Value (expected future value of water)	Market price	Production/ sales
(4)	prob: 0.5 → 150 0.3 → 100 0.2 → 50 WV = 0.5 *150 + 0.3*100 + 0.2*50 = 110	150	100



Results: Optimum strategy (Stage 2-Node 4)



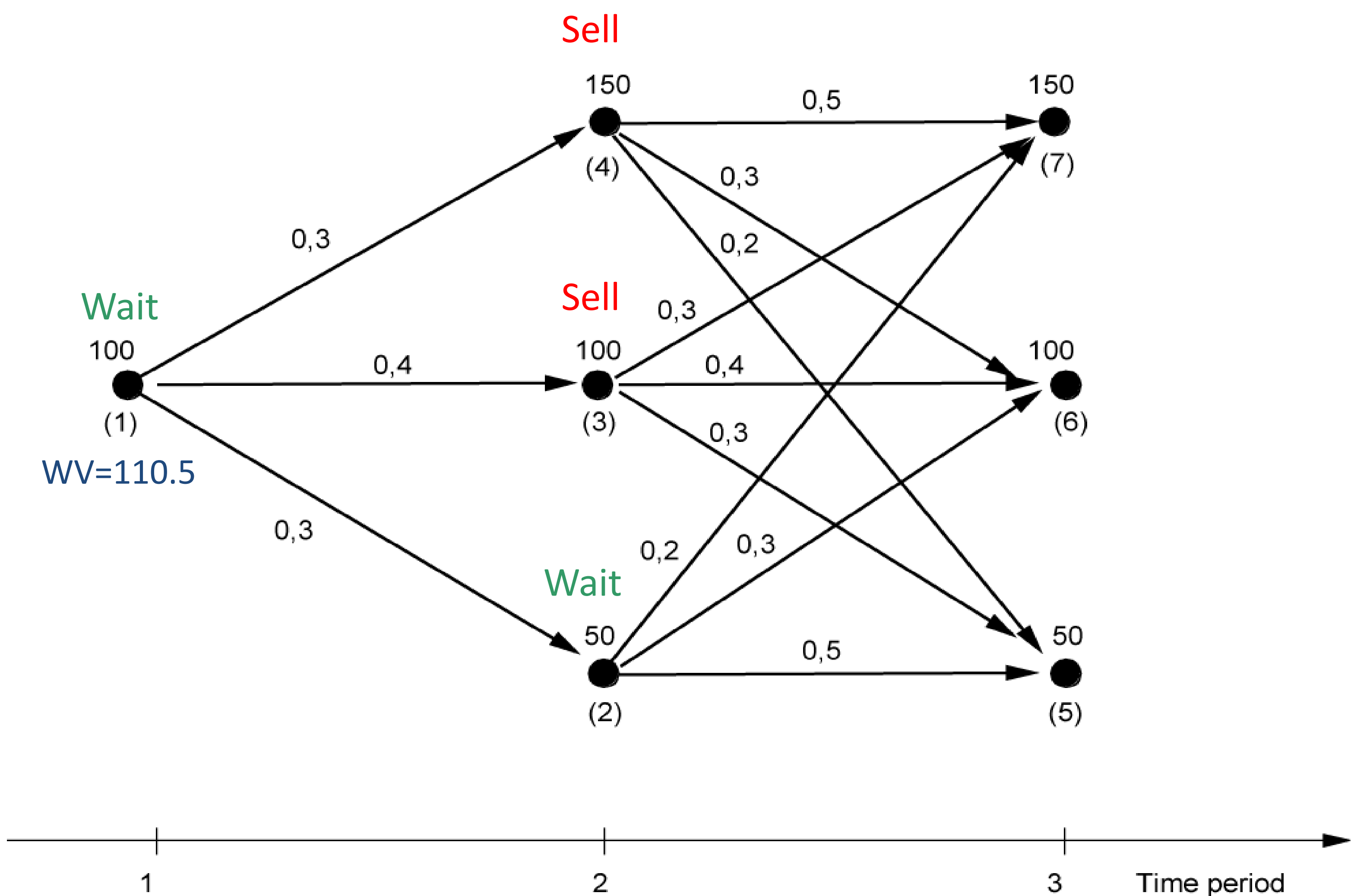
Stage 1-Node 1

Scenario	Price in stage1	Price in stage 2	Price in stage	Probability	Selling price	Expected Income NOK/MWh
1 (Node (1)(2)(5))	100	50	50	0.3 *0.5 = 0.15	50	7.5
2 (Node (1)(2)(6))	100	50	100	0.3 *0.3 = 0.09	100	9
3 (Node (1)(2)(7))	100	50	150	0.3 *0.2 = 0.06	150	9
4 (Node (1)(3)(5))	100	100	50	0.4 *0.3 = 0.12	100	12
5 (Node (1)(3)(6))	100	100	100	0.4 *0.4 = 0.16	100	16
6 (Node (1)(3)(7))	100	100	150	0.4 *0.3 = 0.12	100	12
7 (Node (1)(4)(5))	100	150	50	0.3 *0.2 = 0.06	150	9
8 (Node (1)(4)(6))	100	150	100	0.3 *0.3 = 0.09	150	13.5
9 (Node (1)(4)(7))	100	150	150	0.3 *0.5 = 0.15	150	22.5
Expected price	**100**	**1[illegible]0**	**100**	**Σ = 1.00**		**Σ = 110.5**

Sell

Wait

Results: Optimum strategy (Stage 1-Node 1)



Conclusion

- Hydropower provides flexibility in a weather-dependent system
- Key challenge: optimal use of limited water resources
  - Water has value (opportunity cost)
- Water value drives decisions
  - Produce now vs. store for future
- Constraints create price differences
  - Reservoir, generation, transmission
- Uncertainty requires stochastic models
  - Two-/multi-stage optimization